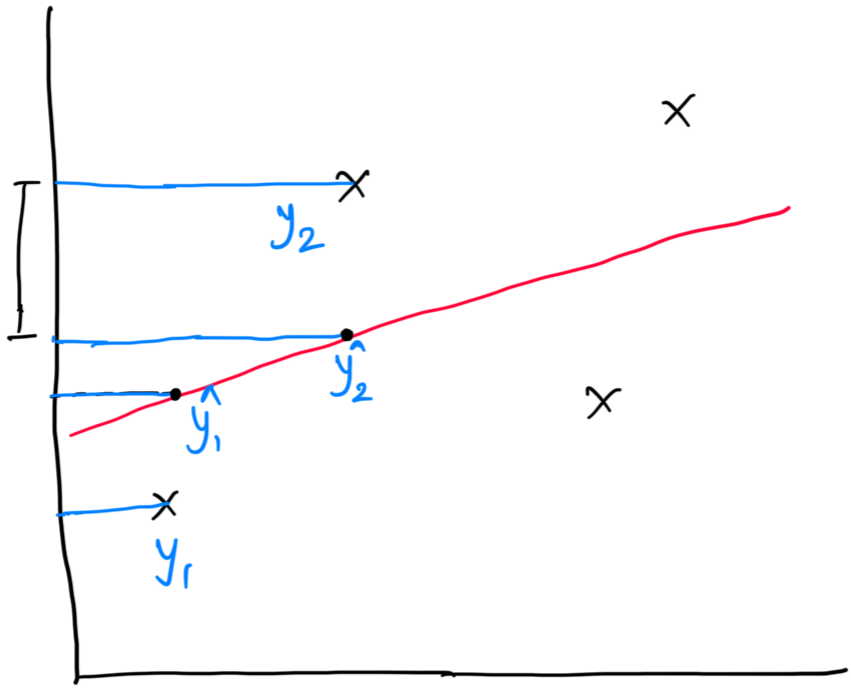
Regression Metrics

1) MAE (Mean Absolute Error)



$$\text{Error} = |y_1 - \hat{y}_1| + |y_2 - \hat{y}_2| + \dots + |y_n - \hat{y}_n|$$

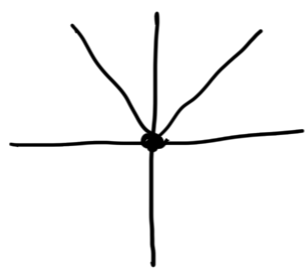
$$\boxed{MAE = \frac{\sum |y_i - \hat{y}_i|}{n}}$$

Advantages

1) Loss has same unit as output
2) Robust to outliers

Disadvantages

1) We are using modulus function to calculate a loss but modulus function is not differentiable at 0 so graph becomes



2) Equal Weighting of Error: MAE penalizes all errors, so large errors are not emphasized.

3) Slow convergence near minimum: Constant gradients can make optimization

unstable or slow near the optimum.

4) Ignores Error direction:
MAE measures only magnitude, not over or under prediction bias.

5) Scale dependent:
Error values depend on the data's units, limiting cross-dataset comparison.

6) Less efficient or Normal data:
MAE is less statistically efficient than MSE for normally distributed errors.

2) MSE - (Mean Squared Error)

$$\boxed{MSE = \sum (y_i - \hat{y}_i)^2}$$

$$\frac{\quad}{n}$$

Advantages

- This is differentiable

Disadvantages

1) Not robust to outliers
2) The unit of the MSE is square of the unit of the target variable.

3) RMSE (Root mean squared Error)

$$RMSE = \sqrt{MSE}$$

$$\boxed{RMSE = \sqrt{\frac{\Sigma (y_i - \hat{y}_i)^2}{n}}}$$

Advantages

1) RMSE output is same as target variable.

Disadvantages

2) Not robust to outliers.

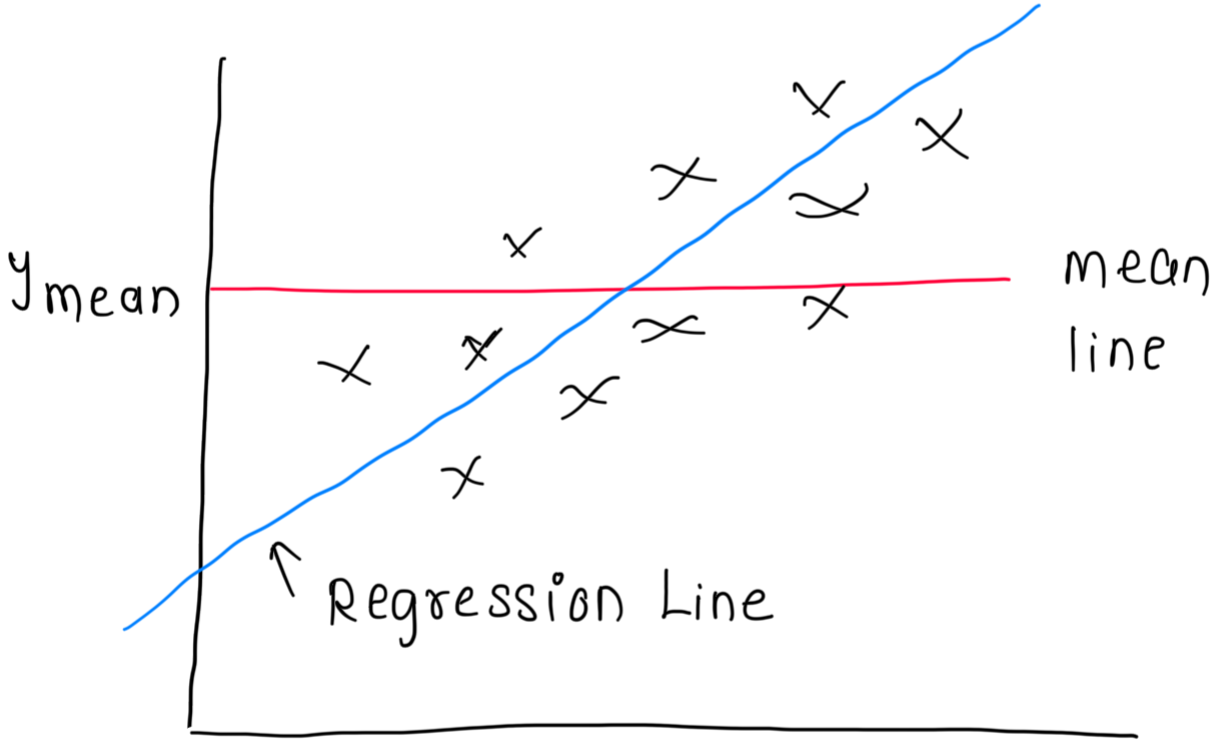
4) R2 score (Coeff of determination or Goodness of fit)

$$\boxed{R2 = 1 - \frac{SS_R}{SS_m}}$$

SS_R = Sum of Squared Error in the regression line

SS_m = Sum of squared Error in the

mean line



$$R2 = 1 - \frac{\left[\sum (y_i - \hat{y}_i)\right]_{Reg}}{\left[\sum (y_i - \hat{y}_i)\right]_{mean}}$$

$R_2 = 0$ when $\dfrac{\left[\sum (y_i - \hat{y}_i)\right]_{Reg}}{\left[\sum (y_i - \hat{y}_i)\right]_{mean}}$ this

term becomes 1

i.e. Regression line = mean line

meaning that both the lines making equal error or regression line becomes mean line.

$R_2 = 1$ when $\dfrac{\left[\sum (y_i - \hat{y}_i)\right]_{Reg}}{\left[\sum (y_i - \hat{y}_i)\right]_{mean}}$ this

term becomes zero. i.e denominator becomes zero meaning that regression line does not create any errors & passes through all points (which is not possible in practice)

$R_2 \rightarrow 1 \Rightarrow$ Good model

$R_2 \rightarrow 0 \Rightarrow$ Bad model

$R_2 < 0$ or negative that means the regression line makes more errors than mean line. i.e. $SS_R > SS_m$.

Interpretation

Suppose R_2 score is 0.8 that means 80% of variance in the output column is being explained by the input column.

5) Adjusted R_2 score

$$\boxed{R^2_{adj} = 1 - \left[\frac{(1-R2)(n-1)}{(n-1-k)}\right]}$$

n - no. of rows

k - no. of independent columns